

Where Does Each Cognitive Capacity Live in the Network?

Residual SAE Feature Relevance Across All 26 Layers of Gemma 3 1B

Early layers

Middle layers

Late layers

Cognitive Task

Sem. Retrieval
Work. Memory
Inhib. Control
Lex. Disambig.
Causal Reason.
Pragmatic Inf.
Theory of Mind
Goal Conflict
Self-Knowledge
Uncertainty

|Feature Relevance|

3.0
2.5
2.0
1.5
1.0
0.5

Layer

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

★

L18

★

L20

★

L16

★

L19

★

L17

★

L21

★

L24

★

L22

★

L15

★

L23